MAXIMILIAN LARTER

I am a plant ecophysiologist and an evolutionary biologist.
My primary research interest is understanding how plants respond and adapt to their environment, in particular in the current climate crisis. I study functional traits to gain insights into key physiological functions, and to predict the impacts of climate change on the performance, survival, and future distributions of wild and cultivated species.

RESEARCH EXPERIENCE

2026 | 2025

Postdoc - modelling urban heat islands and carbon storage of UB

Sylvain Delzon's lab — INRAE, Bordeaux, France

- This partnership with local startup Netcarbon aims to quantify and simulate the impacts of management, development and land use changes on campus. Building a model to estimate soil carbon and crop yield changes based on new agro-ecological farming practices.

2024 | 2021

Postdoc - Plant hydraulics & trait trade-offs, forest ecology and biogeography

Sylvain Delzon's lab — INRAE, Bordeaux, France

- With large trait databases and forest inventory data, we investigated how plant functional traits (embolism and frost resistance) interact and how they shape species distributions and ecology.

2021 | 2019

Postdoc - Herb hydraulics, positive root pressure and drought resistance in Brassicaceae

Frederic Lens's lab — Naturalis, Leiden, The Netherlands

- This project combines classical plant hydraulics adapted to non-woody species with xylem anatomy, micro-CT and modeling to obtain a holistic picture of herb hydraulics during drought. We are notably looking at positive root pressure, which has been hypothesized to aid in recovering from drought by refilling embolised xylem conduits.

2019 | 2017

Postdoc - Evolution of the anthocyanin pathway in Iochrominae

Stacey Smith's lab — University of Colorado, Boulder, USA

- In Iochrominae, several lineages have independently lost floral anthocyanin pigmentation altogether, resulting in white or yellow flowers. We found that the mechanism behind these fixed evolutionary losses is convergent down-expression in three downstream genes of the pigment biosynthetic pathway.

2016 | 2012

PhD - Evolution of cavitation resistance in conifers

Sylvain Delzon's lab — Université de Bordeaux, France

- This thesis expands our understanding of the evolution of vascular plants regarding severe drought. We show that embolism resistance varies 9-fold across over 250 conifer species, thanks to changes in bordered-pit anatomy. We link embolism resistance evolution to increased diversification rate, and describe the remarkable evolution of *Callitris* xylem during the aridification of Australia over the last 30MA.
- *Supervisors*: Sylvain Delzon and Jean-Christophe Domec

CONTACT

maximilian.larter@gmail.com
twitter / Bluesky
maxlarter.github.io
+33 679709275
10 rue HG Wells,
33600 Pessac, France
August 5th, 1987. Derby (UK)

SKILLS

Languages

- Fluent / native speaker in **English** and **French**.

Software, statistics

- R, git, Python, SAS, QGIS
- Phylogenetics
- Modelling

Plant physiology

- plant hydraulics (Cavitron, optical technique)
- plant frost resistance
- gas exchange, sapflow
- wood anatomy, microscopy

Molecular biology, phylogenetics

- DNA/RNA extractions, qPCR
- NGS library prep

Academic service

- Peer reviews: (>20) New Phytol, Tree Phys, PCE, Am J Bot, BMC Evol. Biol.
- Postdoc & short contract rep. CU Biogeco 2021-2022

Last updated on 2026-09-15.

2012 — Research assistant - Evolutionary patterns of cavitation resistance in conifers (6 months)
Supervisor: Sylvain Delzon — Université de Bordeaux, France

2011 — MSc thesis - Convergent evolution of drought tolerance in conifers (6 months)
Supervisor: Sylvain Delzon — Université de Bordeaux, France

2010 — MSc thesis - Population genetics of the olive (*Olea europaea*) complex (2 months)
Supervisor: Guillaume Besnard — Imperial College London, UK

EDUCATION

2025 | 2022 — Qualification CNU
Sections 66 Physiologie & 67 Biol. des populations et écologie

2016 | 2012 — PhD in evolutionary, functional and community ecology.
Thesis: Evolution of cavitation resistance in conifers
Université de Bordeaux — Bordeaux, France
- *Supervisors*: Sylvain Delzon and Jean-Christophe Domec

2011 | 2010 — MSc - Terrestrial Ecosystem Functioning and Modelling
Université de Bordeaux — Bordeaux, France

2008 | 2005 — BSc - Organismal Biology
Université d'Orléans — Orléans, France

PUBLICATIONS

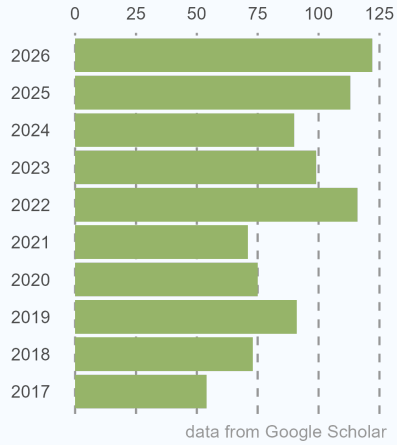
2026

G Bortolami, K Steppe, J Van den Bulcke, **M Larter**, S Balazadeh, F Lens et al. (2026) Positive pressure does not efficiently repair xylem embolism after drought, *Journal of Experimental Botany.* pdf

A Pereira-Zaldívar, G Bortolami, **M Larter**, S Delzon, R Burlett, J van Haasteren, M van Unen, K Wiegant, S Balazadeh, F Lens (2026) Pit Membrane Thickness Determines Differences in Embolism Resistance Among Brassica oleracea Genotypes, *Physiologia Plantarum.* pdf

S Trueba, R Burlett, X Bouteiller, T Burneau, G Forget, **M Larter**, D N'Do, C Ziegler, S Delzon (2026) Ecological drivers and phylogenetic patterns of leaf minimum conductance variability in vascular plants, *New Phytologist.* pdf

965 citations, h-index 13



2025

- **M Larter**, G Charrier, S Delzon, W Hammond, A Baranger, C Bertrand, N Martin-StPaul, G Kunstler (2025) Weak global trade-off between frost and drought resistance in trees, *New Phytologist.* pdf
- L. Wheeler, **M Larter**, SD Smith (2025) MYB regulator of 'colorless' flavonols underlies the evolution of red flowers in Iochroma (Solanaceae), *G3.* pdf

2024

- **M Larter**, A Akhmedov, C Payne, S Delzon, T Klein (2024) High and variable hydraulic resistance to embolism among three Asian juniper species, *Journal of Plant Hydraulics.* 10 (2) pdf
- A Baranger, T Cordonnier, G Charrier, S Delzon, **M Larter**, N Martin-StPaul, G Kunstler (2024) Living on the edge - physiological tolerance to frost and drought explains range limits of 35 European tree species, *Ecography.* pdf
- G Bortolami, TA de Werk, **M Larter**, A Thonglim, F Lens et al. (2024) Integrating Gene Expression Analysis and Ecophysiological Responses to Water Deficit in Leaves of Tomato Plants, *Scientific Reports.* pdf

2023

- Y Song, XP Bouteiller, **M Larter**, C Plomion, F Sin, S Delzon, (2023) A safe breeding ground: genetically improved maritime pine for growth and stem form has more efficient but not more vulnerable xylem, *Tree Physiology.* pdf

2022

- A Thonglim, G Bortolami, S Delzon, **M Larter**, R Offringa, F Lens et al. (2022) Drought response in Arabidopsis displays synergistic coordination between stems and leaves, *Journal of Experimental Botany.* pdf

2021

- A Hooft van Huysduynen, S Janssens, V Merckx, R Vos, L Valente, **M Larter** et al. (2021) Temporal and palaeoclimatic context of the evolution of insular woodiness in the Canary Islands, *Ecology and Evolution* pdf

2020

- A Thonglim, S Delzon, **M Larter**, O Karami, A Rahimi, R Offringa et al. (2020) Intervessel pit membrane thickness best explains variation in embolism resistance amongst stems of Arabidopsis thaliana accessions, *Annals of Botany.* pdf

2019

- **M Larter**, A Dunbar-Wallis, AE Berardi, SD Smith (2019) Developmental control of convergent floral pigmentation across evolutionary timescales, *Developmental dynamics*, 248 (11), 1091-1100. pdf

R Deanna, **M Larter**, GE Barboza, SD Smith (2019) Repeated evolution of a morphological novelty: a phylogenetic analysis of the inflated fruiting calyx in the Physalideae tribe (Solanaceae), *American Journal of Botany*, 106 (2), 270-279. pdf

2018 **M Larter**, A Dunbar-Wallis, AE Berardi, SD Smith (2018) Convergent evolution at the pathway level: predictable regulatory changes during flower color transitions, *Molecular biology and evolution*, 35 (9), 2159-2169. pdf

2017 **M Larter**, S Pfautsch, JC Domec, S Trueba, N Nagalingum, S Delzon (2017) Aridity drove the evolution of extreme embolism resistance and the radiation of conifer genus Callitris, *New Phytologist*., 215 (1), 97-112. pdf

C Sáenz-Romero, **M Larter**, N González-Muñoz, C Wehenkel et al. (2017) Mexican conifers differ in their capacity to face climate change, *Journal of Plant Hydraulics*., 4, e003. pdf

2016 B Castagneyrol, H Jactel, EG Brockerhoff, N Perrette, **M Larter**, S Delzon et al. (2016) Host range expansion is density dependent, *Oecologia*., 182 (3), 779-788. pdf

M Larter (2016) The evolution of cavitation resistance in conifers. *Université de Bordeaux, PhD Thesis* pdf

2015 **M Larter**, TJ Brodribb, S Pfautsch, R Burlett, H Cochard, S Delzon (2015) Extreme aridity pushes trees to their physical limits, *Plant Physiology*, 168 (3), 804-807. pdf

2014 PS Bouche, **M Larter**, JC Domec, R Burlett, P Gasson, S Jansen, S Delzon (2014) A broad survey of hydraulic and mechanical safety in the xylem of conifers, *Journal of Experimental Botany*, 65 (15), 4419-4431. pdf

G Besnard, J Dupuy, **M Larter**, P Cuneo, D Cooke, L Chikhi. (2014) History of the invasive African olive tree in Australia and Hawaii: evidence for sequential bottlenecks and hybridization with the Mediterranean olive, *Evolutionary Applications*, 7 (2), 195-211. pdf

GRANTS AND FUNDING

Active participation in grant writing

ANR projects, Calls for Recherche Inter-disciplinaire et Exploratoire and Prématuration-Innovation at Université de Bordeaux, Fondation pour la Recherche et la Biodiversité IDEASHARE & DATASHARE

2020 ### Alberta Mennega Stichting fieldwork grant

1,250€ Leiden (The Netherlands)

2014 External mobility grant from the COTE Cluster of Excellence
3,000€ Bordeaux (France)

Research Exchange Program (Inbound)
AU$3350 Western Sydney University, NSW (Australia)

OUTREACH AND PRESS (BY ME)

2026 Webinar: Space4Urban Adaptation and Floods - NEREUS and Space for Climate Observatory - view on Youtube

2025 Interview: Végétation, Villes : les acteurs du SCO vous en parlent - Space for Climate Observatory - view on Youtube

Tree physiology workshop - College Olympe de Gouges Cadaujac - Climathon

2016 Three minute thesis - MT180 Final Université de Bordeaux - view on Youtube

2013 **M Larter**, Le Pinetum de Bedgebury: la plus belle collection de conifères du monde, *Jardins de France*. pdf

M Larter, P Bouche, Les conifères, une famille à évolution complexe, *Jardins de France*. pdf

TEACHING EXPERIENCE

2023 | 2021 Lecture (1h) - The evolution of secondary woodiness
MSc course Plant Physiology Université de Bordeaux

2022 Supervision of student project (shared, 6 months)
Climatic tolerance of city trees
Research assistant P. Colombet Université de Bordeaux

Supervision of part time student project (6 months)
Embolism resistance of Mediterranean trees
BSc student C Payne Université de Bordeaux

2021 Supervision of student project (6 months)
Xylem anatomy of embolism resistant Conifer species
MBO student H Hereijgers Hogeschool Inholland Delft

2020 Lecture (30 min) - Functional traits case study
MSc course Methods in Biodiversity Analysis Leiden University

Supervision of student project (6 months)
Response to drought of a giant woody cabbage cross
MSc student J van Haasteren — Leiden University

Supervision of student project (6 months)
Xylem anatomy in relation to embolism resistance in Cupressaceae
BSc student C van Kessel — Leiden University

2012 Practical - plant physiology "lab day"
MSc course Plant Physiology — Université de Bordeaux

CONFERENCES AND PRESENTATIONS

2025 Poster - "Weak global trade-off between frost and drought resistance in trees"
Xylem International Meeting XIM6 — Sevilla (Spain)

2024 Poster - "Global variation and trade-off of drought and frost resistance in trees"
13th International Plant Cold Hardiness Seminar — Clermont-Ferrand (France)

2022 Talk - "Trade off in cold and drought tolerance in trees"
Xylem International Meeting XIM5 — Wurzburg (Germany)

2019 Poster - "Genetic basis of convergent evolution of the anthocyanin pathway and floral pigmentation in Iochrominae"
Society for Integrative and Comparative Biology — Tampa, Florida

2017 Talk - "Linking changes in gene expression to the macroevolution of flower color in Iochrominae (Solanaceae)"
Evolution Meeting — Portland, Oregon

2015 Talk - "Evolution of drought tolerance in conifers - *Callitris* in Australia"
LabEx Day (LabEx COTE) — Bordeaux (France)

Talk - "The evolution of cavitation resistance in conifers and the case of world-record *Callitris*"
Xylem International Meeting XIM2 — Bordeaux (France)

2014 Talk - "The evolution of cavitation resistance in Conifers"
HIE Seminar Series - UWS — Richmond, NSW (Australia)

2012

Poster – “Global variation and evolution of drought tolerance in Conifers”

Journées de la Société Française de Systématique

Paris (France)